

JEREMIAH NYAGAH NATIONAL POLYTECHNIC

Electrical Installation technician Level 6

ENG/OS/EI/CC/01/6

Apply Engineering Mathematics

2 Hours: 2024

SECTION A (40 marks)

Answer ALL the questions section.

1. Solve $e^{x-1} = 2e^{3x-4}$ correct to 4 significant figures. (4marks)
2. Solve the equations by elimination method; (4marks)
 $7x - 2y = 26$
 $6x + 5y = 29$
3. Prove the trigonometric identity (4marks)
 $\frac{1+\cot \theta}{1+\tan \theta} = \cot \theta$
4. a) Solve the equation $\log_x 81 + \log_3 x = 5$ (4 marks)
5. Determine the root of the equation $x^2 - 3x - 4 = 0$ using the completing square method. (4marks)
6. Prove the Identity $\frac{1+\sin \theta}{\cos \theta} = \frac{\cos \theta}{1-\sin \theta}$ [4marks]
7. Solve the equation $(2^{x-5})(5^{x+1}) = 62.5$. [4marks]
8. Solve the equation $\log(x^2 - 3) - \log x = \log 2$. [4marks]
9. Solve the following simultaneous equations using both elimination and substitution. [8marks]
 $x + y + z = 4$
 $2x - 3y + 4z = 33$
 $3x - 2y - 2z = 2$
10. Prove that $\cot 2x + \operatorname{cosec} 2x = \cot x$. [4marks]